

Mrinank Gaur

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SUMMARY

Full-stack software engineer with hands-on experience building production-style systems across React/Next.js, Node.js, FastAPI, and embedded IoT (ESP32), including REST APIs, real-time WebSocket systems, and applied machine learning (Scikit-Learn, LSTM). Strong DSA foundation, with a track record of independently shipping end-to-end software projects.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, Material-UI, OpenCV, Pandas, NumPy, Matplotlib, Scikit-Learn

Databases & Backend: PostgreSQL, MongoDB, Prisma, Convex, JWT Auth, Clerk, WebSockets

Cloud & Tools: AWS, Linux, Git, VS Code, Claude Code, Generative AI tooling

Data Structures & Algorithms: 290+ problems solved on LeetCode (Dynamic Programming, Backtracking, Union-Find, Graphs) - **100-day** solve streak

EXPERIENCE

Maven Silicon

Digital Design Intern

Chennai, India

May 2024 – Jul 2024

- Enabled reliable SoC integration by designing and implementing an APB–AHB bridge in Verilog RTL, verifying functional correctness against bus protocol specifications through simulation, synthesis, and verification in Cadence EDA tools.

PROJECTS

Real-Time Mood Detection via Gait Analysis

Nov 2025 – Apr 2026

Python, FastAPI, Next.js, WebSockets, ESP32, Scikit-Learn | [GitHub](#)

- Achieved **>90%** mood-classification accuracy by building a Random Forest ML pipeline trained on real-time gait data captured from ESP32 force and IMU sensors.
- Delivered low-latency biometric predictions in a live demo by streaming continuous model output from a Python FastAPI server over WebSockets.
- Improved data interpretability for end users by building a responsive Next.js dashboard with live pressure heatmaps, foot-orientation metrics, and real-time streaming views.

Shopalytics — B2B SaaS for Shopify Stores

Aug 2025 – Sep 2025

Next.js, Node.js, Prisma, PostgreSQL | [GitHub](#) | [Live Demo](#)

- Automated store data ingestion and synchronization for Shopify merchants by integrating Shopify APIs into a Next.js/Node.js SaaS platform.
- Supported secure multi-tenant usage across customer accounts by engineering a Prisma + PostgreSQL backend with role-based authentication.
- Increased visibility into business performance for store owners by delivering interactive dashboards tracking revenue, customer, and order trends.

Flex-Sensor Smart Glove — Sign Language Recognition & Speech Translation

Jul 2025 – Nov 2025

ESP32, Python, LSTM, Next.js, MPU6050 | [GitHub](#)

- Achieved **95%** stable sensor calibration and **50–100 Hz** real-time data streaming by building a wireless smart glove with 5 flex sensors and an IMU on ESP32.
- Reached **80.4%** test accuracy across 30+ ASL gestures by training an LSTM gesture-recognition model, achieving **<100ms** end-to-end inference latency.
- Enabled real-time sign-to-speech communication for users by developing a Next.js app with live gesture visualization, 5-language translation, and instant speech output.

Boardly — Collaborative Whiteboard

May 2025 – Jul 2025

Next.js, TypeScript, LiveBlocks, Convex | [GitHub](#) | [Live Demo](#)

- Built a **real-time collaborative whiteboard** with multi-user editing, live cursor tracking, custom rooms, and synchronized updates using LiveBlocks
- Developed a drawing toolkit with **5+ interactive tools** including shapes, text, notes, and freehand drawing, with layer management
- Implemented **authentication, board sharing, and real-time data** using Clerk and Convex, and deployed the application on Vercel

EDUCATION

Vellore Institute of Technology

2022 – 2026

B.Tech, Electronics and Communication Engineering — CGPA: 8.97 / 10

The Khaitan School

Graduated 2022

Class XII (CBSE): 89.2% | Class X (CBSE): 96.2%

LEADERSHIP & EXTRACURRICULAR

Technocrats Robotics, VIT University

Oct 2023 – May 2025

Robotics Engineer

- Enabled precise autonomous navigation by designing and calibrating 8-axis movement control for an autonomous rover, and collaborated on modular software for motion planning and diagnostics.